

No. of Functional Features Included in the Solution

Date	1 October 2026
Team ID	SWTID-2026-1235
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Concept Explanation	Explains a topic in simple and understandable language	Explanation Module	Done	
2	Interactive Q&A	Answers student questions using AI in a concise and educational manner	Q&A Module	Done	
3	Quiz Generation	Generates multiple-choice questions with four options and correct answers	Quiz Module	Done	
4	Text Summarization	Converts lengthy text into simple summaries with key points	Summary Module	Done	
5	Personalized Learning Path	Creates structured learning paths with beginner, intermediate and advanced levels	Learning Path Module	Done	
6	AI Diagram Generation	Generates diagrams from topics using Mermaid, including flowcharts, architecture, sequence diagrams, mind maps, ER and class diagrams	Diagram Module	Done	
7	Google Gemini AI Integration	Uses Gemini for AI-powered explanations, Q&A, quizzes, summaries and learning recommendations	Gemini Client	Done	
8	Local AI Model Support	Uses LaMini-Flan-T5 for local concept explanation with Gemini fallback	Explanation Module	Done	
9	REST API	FastAPI endpoints for Q&A, explanation, summary, quiz, diagram and learning recommendations	FastAPI Backend	Done	
10	Web Interface	HTML/CSS-based interface for interacting with EduGenie	Frontend	Done	
11	Live Frontend-Backend Integration	Connects the web interface with backend AI services through API requests	Integration	Done	
12	Error Handling	Provides structured error responses for invalid input and AI-service failures	Backend	Done	
13	Input Validation	Validates required inputs and limits excessively large requests	Backend	Done	

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
14	Health Check API	Provides an endpoint to verify whether the application is running	Backend	Done	
15	Mermaid Diagram Type Selection	Supports automatic or selected diagram types such as flowchart, architecture, sequence, mindmap, ER, class and state diagrams	Diagram Module	Done	
16	Future Enhancements	Planned improvements to extend the learning assistant	Roadmap	Pending	

Feature Summary

Metric	Count / Value
Total Features Listed	16
Implemented Features	15
Pending Features	1
Core Learning Features	6
AI / Intelligence Features	3
Backend / API Features	4
Frontend / Integration Features	2
Database / Storage Features	0

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Web interface, topic input, question input, quiz interface, result display, summary display, learning-path display, diagram display
2	AI / Learning Features	6	Concept explanation, interactive Q&A, quiz generation, text summarization, personalized learning path, AI diagram generation
3	AI Model Integration	2	Google Gemini AI, local LaMini-Flan-T5 model, Gemini fallback mechanism, AI prompt processing
4	Backend / API	5	FastAPI server, REST API endpoints, input validation, error handling, health-check endpoint, request processing, response generation
5	Integration	1	Frontend-backend integration, API communication, real-time request/response handling, AI service integration
6	Database / Storage	0	No database, stateless application design, environment-based configuration using .env, no persistent user-data storage